

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
/** Use GROUP BY for the distribution across states
    Use COUNT on distinct customer_id for total number of customers **/

-- Sort on customer count in descending order
SELECT c.state,
       COUNT(DISTINCT customer_id) AS customer_count,
       SUM(COUNT(DISTINCT customer_id)) OVER() AS total_customer_count
FROM order_t o
INNER JOIN customer_t c USING(customer_id)
GROUP BY c.state
ORDER BY customer_count DESC;
```

Output:

state	customer_count	total_customer_count
California	97	994
Texas	97	994
Florida	86	994
New York	69	994
District of Columbia	35	994
Colorado	33	994
Ohio	33	994
Alabama	29	994
Washington	28	994
Arizona	26	994
Illinois	25	994
Pennsylvania	25	994
Virginia	24	994

NOTE: The first screenshot is for descending order and the second output is for ascending order (The query pasted above is for descending order and the below query sorts in ascending order)

```
-- Sort on customer count in ascending order
SELECT c.state,
       COUNT(DISTINCT customer_id) AS customer_count,
       SUM(COUNT(DISTINCT customer_id)) OVER() AS total_customer_count
FROM order_t o
INNER JOIN customer_t c USING(customer_id)
GROUP BY c.state
ORDER BY customer_count ASC;
```

state	customer_count	total_customer_count
Maine	1	994
Vermont	1	994
Wyoming	1	994
Mississippi	2	994
North Dakota	2	994
New Hampshire	3	994
Montana	3	994
New Mexico	5	994
Arkansas	6	994
Delaware	6	994
Hawaii	6	994
Idaho	7	994
Nebraska	7	994

Observations and Insights:

- A total of **994** customers have placed orders with New Wheels.
- A significant number of orders originated from states such as California, Texas, Florida, and New York
- Fewer orders were received from states like Maine, Vermont, Wyoming, and Mississippi.
- The above query returned 49 rows

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
/** GROUP BY vehicle_maker
    SORT BY customer_count in descending order
    LIMIT for 5 records **/

SELECT vehicle_maker,
       COUNT(DISTINCT customer_id) AS customer_count
FROM order_t o
INNER JOIN product_t USING(product_id)
INNER JOIN customer_t USING(customer_id)
GROUP BY vehicle_maker
ORDER BY customer_count DESC
LIMIT 5;
```

Output:

	vehicle_maker	customer_count
▶	Chevrolet	83
	Ford	63
	Toyota	52
	Dodge	50
	Pontiac	50

Observations and Insights:

- The top 5 performing vehicle makers are Chevrolet, Ford, Toyota, Dodge, Pontiac
- The query without limit clause resulted 54 records indicating the total vehicle makers are 54 in number.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

/** Using the subquery the vehicle makers are ranked using windows function 'rank' grouping by state and vehicle maker. In the outer query filter only the rank 1 vehicle makers for each state**/

```
SELECT vm_ranked.*
FROM
    ( SELECT state,
            vehicle_maker,
            COUNT(DISTINCT customer_id) AS customer_count,
            RANK() OVER(PARTITION BY state ORDER BY COUNT(DISTINCT
customer_id) DESC) AS vehicle_maker_rank
    FROM order_t o
    INNER JOIN product_t USING(product_id)
    INNER JOIN customer_t USING(customer_id)
    GROUP BY state, vehicle_maker
    ORDER BY state
    ) AS vm_ranked
WHERE vm_ranked.vehicle_maker_rank=1
ORDER BY customer_count DESC;
```

Output:

state	vehide_maker	customer_count	vehide_maker_rank
Texas	Chevrolet	9	1
Florida	Toyota	7	1
California	Audi	6	1
California	Chevrolet	6	1
California	Dodge	6	1
California	Ford	6	1
California	Nissan	6	1
Ohio	Chevrolet	6	1
Alabama	Dodge	5	1
Colorado	Chevrolet	5	1
Maryland	Ford	5	1
New York	Toyota	5	1
New York	Pontiac	5	1
Virginia	Ford	5	1
Washing...	Chevrolet	5	1
District o...	Chevrolet	4	1
Indiana	Mazda	4	1
Missouri	Chevrolet	4	1
Arizona	Cadillac	3	1
Arizona	Pontiac	3	1
Georgia	Toyota	3	1

Observations and Insights:

- In Texas and Florida, Chevrolet and Toyota are the leading vehicle makers, respectively.
- In California, multiple vehicle makers are closely competing for the top position. There are some states where multiple vehicle makers are sharing the top rank.
- The above query produced 143 rows.

Question 4: Find the overall average rating given by the customers.

What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
/** Using CASE the customer rating is converted to number and then using  
aggregate function average rating is found in the subquery. The results are  
queried in outer query and then window function AVG is used for total average  
customer rating **/
```

```
SELECT quarter_number,  
       round(avg_customer_rating,2) as avg_customer_rating,  
       round(AVG(avg_customer_rating) OVER(),2) AS tot_customer_rating  
FROM  
      (SELECT quarter_number,  
              AVG(CASE  
                  WHEN customer_feedback = 'Very Bad' THEN 1  
                  WHEN customer_feedback = 'Bad' THEN 2  
                  WHEN customer_feedback = 'Okay' THEN 3  
                  WHEN customer_feedback = 'Good' THEN 4  
                  WHEN customer_feedback = 'Very Good' THEN 5  
                  ELSE 0  
              END) AS avg_customer_rating  
      FROM  
          order_t  
      GROUP BY quarter_number  
      ORDER BY quarter_number) as q
```

Output:

quarter_number	round(avg_customer_rating,2)	tot_customer_rating
1	3.55	3.07
2	3.36	3.07
3	2.96	3.07
4	2.40	3.07

Observations and Insights:

- The overall average customer rating stands at 3.07, which corresponds to "Okay" rating
- Customer ratings have shown a downward trend across successive quarters

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
/** Calculate the percentage of each feedback type by using conditional
aggregation. For each feedback category, use a CASE statement to count the
occurrences and then divide by the total count of feedback for the quarter,
multiplied by 100 to get the percentage**/

SELECT quarter_number,
       ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad'
                     THEN 1 ELSE 0 END)*100/count(*),2) AS very_bad_percentage,
       ROUND(SUM(CASE WHEN customer_feedback = 'Bad'
                     THEN 1 ELSE 0 END)*100/count(*),2) AS bad_percentage,
       ROUND(SUM(CASE WHEN customer_feedback = 'Okay'
                     THEN 1 ELSE 0 END)*100/count(*),2) AS okay_percentage,
       ROUND(SUM(CASE WHEN customer_feedback = 'Good'
                     THEN 1 ELSE 0 END)*100/count(*),2) AS good_percentage,
       ROUND(SUM(CASE WHEN customer_feedback = 'Very Good'
                     THEN 1 ELSE 0 END)*100/count(*),2) AS very_good_percentage
FROM   order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

quarter_number	very_bad_percentage	bad_percentage	okay_percentage	good_percentage	very_good_percentage
1	10.97	11.29	19.03	28.71	30.00
2	14.89	14.12	20.23	22.14	28.63
3	17.90	22.71	21.83	20.96	16.59
4	30.65	29.15	20.10	10.05	10.05

Observations and Insights:

- The percentages of "Very Good" and "Good" ratings have declined steadily over the quarters
- The percentage of "Okay" ratings increased during the first three quarters but dropped in the last
- "Bad" and "Very Bad" ratings have been on an upward trend throughout the quarters.
- These statistics indicate a consistent decline in customer satisfaction over time.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
/**Count the number of orders for each quarter**/  
  
SELECT quarter_number,  
       COUNT(order_id) AS order_count  
FROM order_t  
GROUP BY quarter_number  
ORDER BY quarter_number;
```

Output:

quarter_number	order_count
1	310
2	262
3	229
4	199

Observations and Insights:

- The number of orders has been declining steadily across the quarters.
- Order volume peaked in Quarter 1 and reached its lowest in Quarter 4.

Question 7: Calculate the net revenue generated by the company.

What is the quarter-over-quarter % change in net revenue?

Solution Query:

/**Net Revenue is the amount obtained by multiplying the number of units sold by the price after deducting the discounts applied.

Quarter over Quarter percentage change in revenue means what is the change in revenue from the subsequent quarter to the previous quarter in percentage.

Calculate the revenue for each quarter by summing the quantity of product and the discounted vehicle price. Use the LAG function to get the revenue from the previous quarter, and then compute the quarter-over-quarter percentage change based on the current and previous revenue values**/

```
SELECT quarter_number,
       ROUND(SUM(SUM((vehicle_price - discount*vehicle_price)*quantity))
OVER(),2) AS tot_revenue,
       ROUND(SUM((vehicle_price - discount*vehicle_price)*quantity),2) AS
net_revenue,
       ROUND(LAG(SUM((vehicle_price - discount*vehicle_price)*quantity))
OVER(ORDER BY quarter_number),2) AS prev_net_revenue,
       ROUND((SUM((vehicle_price - discount*vehicle_price)*quantity) -
       LAG(SUM((vehicle_price - discount*vehicle_price)*quantity)) OVER(ORDER BY
quarter_number) ) * 100 /
       SUM((vehicle_price - discount*vehicle_price)*quantity),2) AS
quarter_over_quarter_percent
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

quarter_number	tot_revenue	net_revenue	prev_net_revenue	quarter_over_quarter_percent
1	48610993.78	18032549.90	NULL	NULL
2	48610993.78	13122995.76	18032549.90	-37.41
3	48610993.78	8882298.84	13122995.76	-47.74
4	48610993.78	8573149.28	8882298.84	-3.61

Observations and Insights:

- The company has generated a total revenue of \$48,610,993.78.
- Net revenue has been declining consistently over the quarters.
- The quarter-over-quarter revenue percentage change is negative, indicating a downward trend.
- The steepest quarter-over-quarter decline occurred in Quarter 3.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
/** sum of net revenue and count the number of orders for each quarter**/  
  
SELECT quarter_number,  
       ROUND(SUM((vehicle_price - discount*vehicle_price)*quantity),2)  
       AS net_revenue,  
       COUNT(order_id) AS order_count  
FROM order_t  
GROUP BY quarter_number  
ORDER BY quarter_number;
```

Output:

quarter_number	net_revenue	order_count
1	18032549.90	310
2	13122995.76	262
3	8882298.84	229
4	8573149.28	199

Observations and Insights:

- Net revenue has been decreasing consistently from one quarter to the next.
- The number of orders also exhibits a downward trend across consecutive quarters.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
/** Compute the average of discount for each credit card type
    Use windows function avg to compute total average on all the card types**/

SELECT credit_card_type,
       ROUND(AVG(discount),2) AS avg_discount_card,
       ROUND(AVG(AVG(discount)) OVER(),2) AS avg_discount
FROM order_t
INNER JOIN customer_t USING(customer_id)
GROUP BY credit_card_type
ORDER BY avg_discount_card DESC;
```

Output:

credit_card_type	avg_discount_card	avg_discount
laser	0.64	0.61
mastercard	0.63	0.61
visa-electron	0.62	0.61
china-unionpay	0.62	0.61
americanexpress	0.62	0.61
maestro	0.62	0.61
instapayment	0.62	0.61
jcb	0.61	0.61
switch	0.61	0.61
diners-club-carte-blanche	0.61	0.61
bankcard	0.61	0.61
diners-club-us-ca	0.61	0.61
diners-club-enroute	0.60	0.61
visa	0.60	0.61
solo	0.59	0.61
diners-club-international	0.58	0.61

Observations and Insights:

- Laser credit cards offer the highest discounts, followed by MasterCard.
- The diners club international credit card is offering lowest discount.
- A total of 16 different credit card types have been used
- The average discount on all card types is about 61 percentage.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
/** Compute the average time taken from the difference of ship date and order
date */

SELECT quarter_number,
       CEIL(AVG(DATEDIFF( ship_date, order_date))) AS avg_time_in_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

quarter_number	avg_time_in_days
1	58
2	72
3	118
4	175

Observations and Insights:

- The average shipping time has been rising progressively each quarter.
- Quarter 4 recorded the highest average shipping time.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
\$ 48,610,993.78	1000	994	3.13 => Okay
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
\$ 8,573,149.28	199	98	*21.50

- The company has generated a total revenue of **\$48,610,993.78**
- A total of **1,000** orders were received over the four quarters.
- The customer base comprises **994** unique customers.
- The company's average customer rating falls under the "**Okay**" category.
- Net revenue has been declining, with the last quarter generating **\$8,573,149.28**.
- The number of orders has been decreasing quarter over quarter, dropping to **199** in the final quarter
- The average time taken to ship is about **98** days.
- *"Good" feedback accounts for **21.50%** and "Very Good" feedback for **22.60%**, combining to a total of **44.10%** positive customer feedback

Business Recommendations

- Investigate the root causes behind the decline in customer ratings across quarters.
- Prioritize service quality enhancements, especially in areas with rising "Bad" and "Very Bad" feedback.
- Launch targeted loyalty or feedback programs to regain customer trust.
- Optimize the supply chain process to improve the average shipping time.
- Evaluate warehouse operations, logistics partnerships, and inventory management to reduce delays.
- Review discount effectiveness by card type and consider adjusting offerings based on performance.
- Analyze any additional reasons for the decrease in the number of orders.
- Revise the pricing strategy and cost control measures to optimize the profitability.